Supplementary Tables

Title: Bilingual Retrieval-Augmented Access to ANVISA Regulatory Documents

Prepared: 2026-07-20

Supplementary Table S1. Expanded development decision table

Question category or system issue	What the system did well	What went wrong	What was changed	What this means
Source cleaning and document identity	The project identified the intended regulatory corpus.	Local source files included formatting artifacts and one misleading local file name.	Source text was cleaned and document identity was checked against official regulation names.	Regulatory RAG needs a stable and auditable corpus before retrieval metrics are interpretable.
Direct article questions	Retrieved and cited the correct article for compact questions.	Some answers still omitted conditions.	Article-level indexing and completeness prompting were added.	Useful for navigation, but citations must be checked.
English questions over Portuguese sources	Improved after Portuguese retrieval anchors.	English completeness was initially weak.	Conservative bilingual term anchors were added.	Bilingual search is feasible but needs controlled terminology.
Rule-plus-exception questions	Often found the right document.	Sometimes gave the exception without the general rule.	Rule-plus-exception failures were tracked separately.	Regulatory answers must preserve legal structure.
Long-list questions	Often found the right article.	Omitted one element, condition, or deadline.	Answer-completeness checks were added.	Average metrics can hide operationally important omissions.
False-premise questions	Could correct the premise and cite evidence.	Automated labels could confuse false premise with refusal.	False-premise questions were kept as a separate category.	These questions should not be merged with ordinary unanswerable questions.
Current or operational facts outside corpus	Refused after calibration.	Earlier system answered from nearby general text.	A stricter refusal rule and separate refusal test were added.	Refusal calibration is central for regulatory safety.
AI-authored prospective-test questions	Confirmed strong answerable-question behavior.	Refusal correctness dropped to 0.700.	Refusal was repaired using separate development data.	AI-authored prospective testing changed the project story.

Supplementary Table S2. Expanded representative answer outcomes

Question type	Example question	System behavior	Interpretation
Correct answer, pharmacovigilance	Which minimum information makes a notification valid and reportable?	Retrieved RDC 406/2020, Art. 32 and listed the minimum elements.	Correct source and correct main answer.
Correct false-premise handling	Is a notification valid without patient identification if the suspected medicine is named?	Answered no and cited RDC 406/2020, Art. 32.	Useful because it corrected the premise.
Correct clinical-trial safety answer	Does possible SUSAR reporting allow unblinding all participants?	Answered no; unblinding is limited to the affected participant and cited RDC 945/2024, Art. 67.	Correct and relevant to trial conduct.

Question type	Example question	System behavior	Interpretation
Correct refusal	What is the current MedDRA version required today?	Refused because the indexed documents mention MedDRA but not the current required version.	Appropriate refusal for current operational information.
Correct refusal	What is the current phone number for Anvisa support for the adverse-event notification system?	Refused because the corpus does not contain current support contact information.	Appropriate evidence boundary.
Incomplete answer	What information must appear for non-printed instructions for use?	Retrieved RDC 751/2022, Art. 51 but omitted that instructions must remain available while the product is on the market.	Correct article, incomplete answer.
Incomplete answer	Can a foreign-manufactured biological product be registered without registration in the manufacturing country?	Gave the exception but did not clearly state the general rule.	Risk of treating an exception as the whole rule.
False refusal	Under RDC 945/2024, until when must DSURs be sent to Anvisa and what is the annual reference date?	Refused in one English prospective-test row although the corpus contained the answer.	Retrieval or answer threshold failure.
Refusal failure before calibration	What is the current electronic form model for notifying SUSARs to Anvisa?	Earlier AI-authored prospective testing sometimes answered with the general electronic-notification requirement.	Related context was mistaken for sufficient evidence.

Supplementary Table S3. Evaluation-stage summary

Stage	Main finding	Use in interpretation
Initial held-out evaluation	English completeness was the weak point; some answers omitted list items, conditions, or exceptions.	Development diagnosis.
Post-repair internal evaluation	High internal scores showed that the repaired system passed the curated benchmark.	Internal technical check, not independent validation.
AI-authored prospective stress test	Answerable questions were mostly supported, but refusal behavior was weak.	Diagnostic prospective evidence that changed the project focus.
Refusal-focused AI-authored test	Refusal improved after separate calibration, with no over-refusal on answerable controls.	Focused AI-authored check of refusal calibration.

Supplementary Table S4. Main-table disposition after revision

Original table	Revision decision	Reason
Table 1. Regulatory domains represented in the corpus	Replaced with a shorter corpus-and-use-case table.	Reduces page splitting and overlaps less with Figure 2.
Table 2. Question categories and corrective actions	Condensed in the main text and expanded as Supplementary Table S1.	Full version was too dense for the main manuscript.
Table 3. Representative examples	Condensed in the main text and expanded as Supplementary Table S2.	Keeps the story visible while moving audit-level detail to the supplement.
Table 4. Summary of evaluation stages	Removed from the main text and moved to Supplementary Table S3.	Redundant with Figure 3A and Figure 3B.

Supplementary Table S5. Decision trace from course notebooks to final ANVISA RAG

Stage	Local evidence	Main decision	Reason	Effect on final system
Course notebook 1: local LLM	Course notebook file and repository copy `notebooks/1_LLM.ipynb`	Start from a Python project pattern with environment.yml, .env, src/config.py, src/model_loader.py, src/engine.py, and main.py.	The course notebook provided a simple, reproducible local LLM scaffold.	The final project kept the separated configuration, model-loading, engine, and app structure.
Course notebook 2: chatbot	Course notebook file and repository copy `notebooks/2_chatbot.ipynb`	Add a chat interface and system prompt rather than a one-shot completion loop.	The target use case was question answering by regulatory users.	The final app kept an interactive Streamlit question-answer interface, although the final evaluation used scripted runs.
Course notebook 3: generic RAG	Course notebook file and repository copy `notebooks/3_rag.ipynb`	Replace the demonstration corpus with selected ANVISA text files.	The project idea was to adapt the teaching RAG workflow from a generic text example to regulatory documents.	The data folder became a closed ANVISA corpus rather than an open web or general-document system.
Borrowed Shenia source	src_RAG_Shenia/engine.py in the recovered project	Test a stronger RAG pattern using semantic splitting, HyDE, reranking, and chat memory.	The borrowed source worked in another project and showed useful retrieval options beyond the teacher notebook.	Some ideas were retained conceptually, especially reranking, but HyDE/semantic splitting/chat memory were not the final regulatory design.
First ANVISA implementation	Recovered ANVISA RAG project folder and Codex recovered copy	Build an ANVISA Streamlit app with LlamaIndex, local embeddings, Gemini, and indexed ANVISA text files.	The original DSAI2 project folder was later moved; the recovered project contains the app, ANVISA data, cleaned_data, evaluation notebooks, and vector stores.	This is the local ancestor of the publication repository.
Environment failure	Early app traceback: ModuleNotFoundError for llama_index	Treat dependency setup as part of reproducibility rather than an incidental app problem.	The app could not run without the correct conda/Python environment and LlamaIndex packages.	Methods now report Python version, packages, and implementation files.
Corpus cleaning	scripts/clean_regulatory_texts.py and reports/cleaned_data_audit.csv	Create raw-to-cleaned source processing before indexing.	Regulatory text exported from web/PDF-like sources contained markup and boilerplate.	The final corpus is indexed from cleaned text, with source links and checksums supplied for reproducibility.
Article-aware indexing	src/article_chunking.py and reports/article_chunk_audit.json	Use one article per chunk when possible, with document and article metadata.	Regulatory questions usually map to articles, not arbitrary token windows.	The final index contains 382 detected articles and 481 chunks.
Hybrid retrieval	src/retrieval.py	Combine dense retrieval and BM25, then merge candidates by reciprocal rank fusion.	Exact terms such as RDC 945/2024, DDCM, DEEC, Art. 33, and IN 338 matter in legal/regulatory text.	Dense and keyword retrieval are both reported and auditable.

Stage	Local evidence	Main decision	Reason	Effect on final system
Reranking	src/retrieval.py and src/config.py	Retrieve more candidates and rerank down to the final answer context.	Early retrieval could find broadly related but not exact passages.	The final answer context uses top reranked passages rather than only raw vector similarity.
English questions over Portuguese sources	src/model_loader.py and evaluation/compare_rewrite_runs.py	Add conservative Portuguese retrieval anchors while preserving regulatory terms.	English completeness was weak in the first held-out evaluation.	Bilingual access is supported, but language-specific metrics are reported separately.
Answer rules	src/prompts.py and evaluation/answer_quality.py	Require article citations and refusal when retrieved evidence is insufficient.	Correct source retrieval did not guarantee complete or safe answers.	Citation correctness, completeness, and refusal correctness became primary endpoints.
AI-judge evaluation	evaluation/publication_eval.py and Supplementary Methods	Use an automated judge because two independent experts were not available.	The project was performed by one researcher.	The claim is limited to technical evaluation; AI judging is not presented as expert validation.
Stress testing after high internal scores	prospective-test output files	Run an AI-authored prospective stress test after internal metrics became high.	Perfect-looking internal scores could reflect saturation, targeted repairs, or same-model-family judging.	The stress test exposed refusal weakness and changed the project interpretation.
Refusal calibration	refusal-development files and refusal-focused AI-authored test	Repair refusal on separate development data, then test a new refusal-focused set.	The first prospective test showed that nearby related text could trigger unsupported answers.	Refusal correctness improved, but answerable-control completeness showed a trade-off.

Supplementary Table S6. Investigation of saturated internal metrics

Observation	Why it was suspicious	Investigation performed	Interpretation for reliability
Internal refusal correctness reached 1.000 on curated unanswerable rows.	The rows were part of a benchmark built during the project and evaluated after targeted repair.	A separate AI-authored prospective stress test was run once after the pipeline was frozen.	The perfect internal value was not treated as proof of general refusal reliability.
Several post-repair means approached 1.000.	The repair sets were not paired repeats of the same baseline set and were created after error review.	Figure 3A labels these as development-stage results, not statistical before-after proof.	The values document that specific failure modes were repaired under controlled conditions.
Prospective answerable rows still showed faithfulness of 1.000.	Faithfulness can saturate when answers are short and tightly tied to retrieved articles.	Completeness, citation correctness, source MRR, article Recall@5, and examples were reported beside faithfulness.	Faithfulness alone is insufficient; incomplete answers remain possible even with supported claims.
The first prospective stress test had refusal correctness of 0.700.	This contradicted the internal refusal value of 1.000.	Failure review showed answers based on nearby general rules when the exact current or operational fact was absent.	This was the main evidence that internal 1.000 metrics were inflated or at least too narrow.
The refusal-focused AI-authored test improved refusal correctness to 0.944.	A high post-calibration score can also be overinterpreted.	Answerable controls were included and their completeness was reported at 0.908.	The final claim is bounded: refusal improved, but completeness trade-offs and closed-corpus limits remain.
The answer model and judge used the same model family.	Same-model-family grading can overestimate quality.	The manuscript reports this as a limitation and provides per-question outputs for audit.	The AI judge supports reproducibility, not independent expert validation.